



04

DML

Data Manipulation Language

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Types SQL Commands

```
graph TD; Root[Types SQL Commands] --> DQL[DQL Data Query Language]; Root --> DDL[DDL Data Definition Language]; Root --> DML[DML Data Manipulation Language]; DQL --> SELECT[SELECT]; DDL --> CREATE[CREATE]; DDL --> DROP[DROP]; DDL --> ALTER[ALTER]; DML --> INSERT[INSERT]; DML --> UPDATE[UPDATE]; DML --> DELETE[DELETE];
```

DQL

Data Query
Language

SELECT

DDL

Data Definition
Language

CREATE

DROP

ALTER

DML

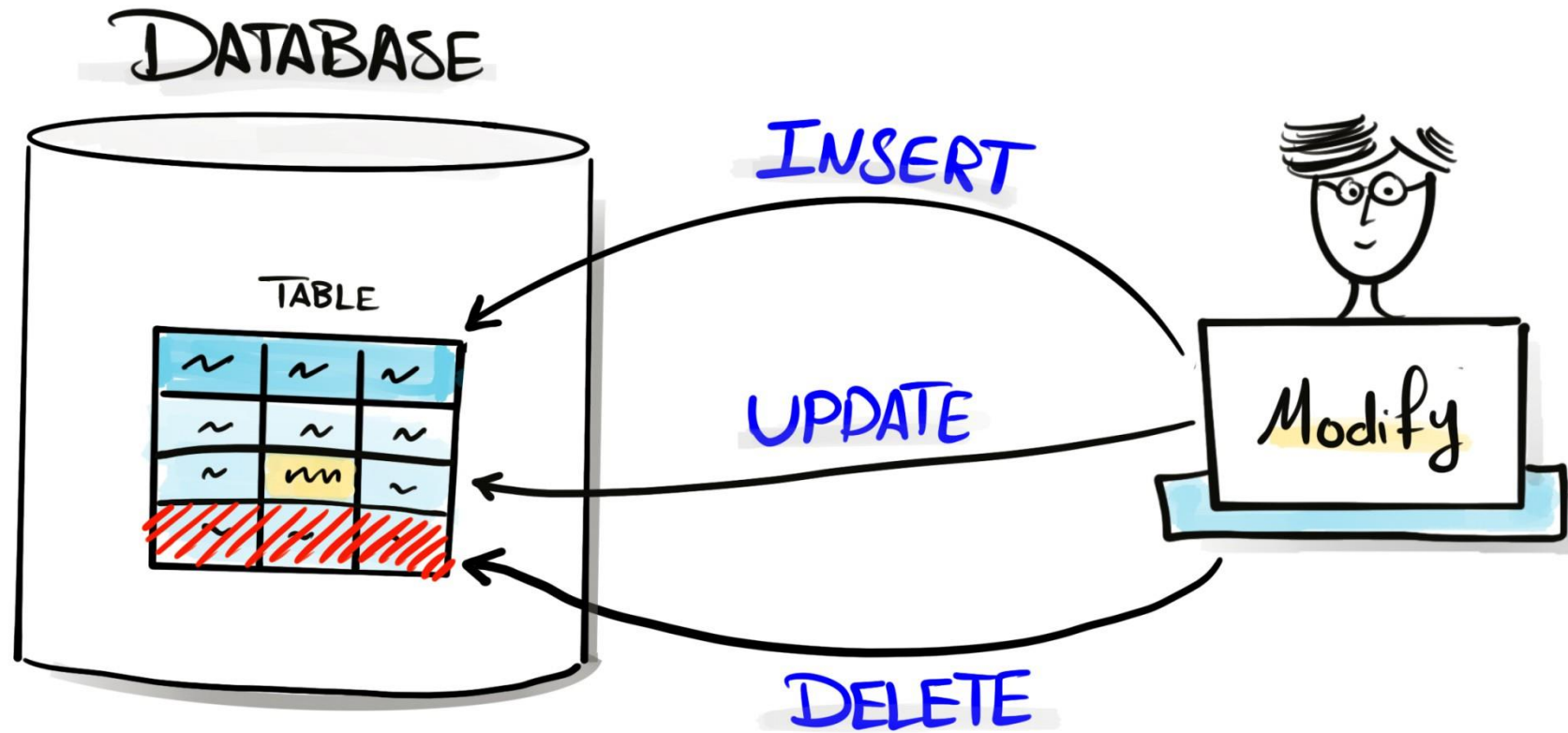
Data Manipulation
Language

INSERT

UPDATE

DELETE

Modify (Manipulate) Your Data



① Manual Entry (Values)

Target Table

~	~	~
~	~	~
~	~	~

Values
~ ~ ~

INSERT
VALUES



Target Table

~	~	~
~	~	~
~	~	~

Query
①
SELECT
FROM
WHERE

Source Table

~	~	~
~	~	~
~	~	~

INSERT
②

Result
~ ~ ~
~ ~ ~
~ ~ ~

② INSERT Using SELECT

INSERT Syntax

OPTIONAL: If **no columns** are specified, SQL expects **values for all columns**

INSERT INTO table_name (column1, column2, column3,...)

VALUES (value1, value2, value3,...)

, (value1, value2, value3,...) ← Multiple Inserts

Rule

- Match the number of **columns** and **Values**.

UPDATE Syntax

```
UPDATE table_name  
    SET column1 = value1,  
        column2 = value2  
    WHERE <condition>
```

NOTE

- Always use WHERE to avoid **UPDATING all rows** unintentionally

DELETE Syntax

```
DELETE FROM table_name  
WHERE <condition>
```

NOTE

- Always use WHERE to avoid DELETING all rows unintentionally